Claude Plasmids Epics

Centralized Space to start cataloging details about our project goals and how we will achieve them.

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Epic 1: Sequence Intelligence

Make the system reliable at understanding and interpreting raw DNA sequences without depending on pre-annotated metadata.

Feature	Description	Priority	Assignee	Status	Notes
MCS Detection	Detect MCS from sequence alone, without relying on library metadata	High	Mialy De…	Awaiting Review	

Orientation Detection	Detect promoter-relative orientation of MCS; handle reverse/CCW promoter configurations	High	Mialy De…	Awaiting Review	
Rich Annotation	Full plasmid annotation beyond the insert (promoters, resistance genes, origins, tags) using pLannotate or equivalent	High	Mialy De…	PR Open	
Gene Extraction from Plasmids	Successfully extract CDS from retrieved plasmid sequences as a fallback when direct NCBI retrieval fails Additionally: Often a user may want to use a particular gene variant that is only available in a plasmid that has been deposited. The program is not able to currently extract the gene from the plasmid. In a previous PR we have implemented Plannotate. Can use plannotate to annotate a plasmid we need to extract a gene from, and use the coordinates to extract the gene of interest.	High	Mialy De…		Supports more complex Phase 1 (and beyond) requests. Relies on annotations.
Codon Optimization	Detect when it should happen then apply tool for optimization.		Person		

Epic 2: External Database Integration

Ensure reliable, up-to-date access to Addgene and NCBI so users aren't blocked by retrieval failures.

Feature	Description	Priority	Assignee	Status	Note

Addgene API Integration	Implement per-user Addgene API key support for live, single-call plasmid retrieval	High	Mialy De…	BLOCKED	Awaiting being allowed to accept the new API Key License Agreement. Messaged Gabi and awaiting response.
Insert Retrieval Reliability	Fix hallucination and incorrect sequence retrieval from NCBI; improve error detection when wrong sequence is returned	Bug			
SnapGene Compatibility	Export constructs in SnapGene-compatible format	Medium			

Epic 3: Construct Design Intelligence

Elevate the tool from sequence assembler to intelligent design partner.

Feature	Description	Priority	Who	Status	Note

Fusion protein support	Basic fusion protein support. Have not tested full scope. May result in more feature requests.	High	MD	Done	Users can select if they want an ATG for internal fusions.
Support tagging plasmids	Some plasmids are designed specifically to allow flexible tagging. They contain everything that is needed for the user to insert at specific points, but do not contain a MCS. This poses a unique challenge for the program to determine where to insert the gene. It needs a different definition method (annotations or literature?) to figure out where to place the insert. This is outside the scope of the MCS PR.			Backlog	Do not have a specific use case for this right now. Can prioritize later
Support Nuclear Targeting Constructs	Allow the user to design a construct with an NLS.	Low		In progress	There is some functionality available but it is not robust.
Support Cytoplasmic Targeting Constructs	Allow the user to design a construct with a cytoplasm targeting sequence	Low			
Design Confidence Score	Predict anticipated construct success — folding, aggregation, expression, cryptic splice sites, etc.	Low			Link to Document
Smart Fusion Protein Design	Identify optimal insertion sites for tags/fusions based on protein structure; avoid functional domain disruption	Low			

Smart Mutation Design	Support gain-of-function and loss-of-function mutation design into target sequences	Low			
Bespoke Promoter Support	Detect when a user requests a custom/non-standard promoter; enter research mode or accept user-supplied sequence	Low			
Modular Parts Swapping	Allow antibiotic resistance and promoter swapping without requiring full backbone change	Medium			
Multi-Output Design	Generate multiple construct variants in a single run (e.g., N-term vs C-term tag placements) with comparative scoring	Low			
Parts Library Support	Cell Science would like to be able to work from a many Parts Library (Promoter, insert, terminator etc). The prompt would be very clear, this should be very straightforward.	High			Pop said they make ~20 plasmids at a time.
Lenti Support	Use validations list to build out lenti support through prompting and additional tooling				
Crispr Support	Use validations list to build out CRISPR support through prompting and additional tooling				

Epic 4: Visualization & Import/Export

Give users a clear, publishable view of what they've designed.

Feature	Description	Priority	Assignee	Status	Note
Plasmid Visualization	Circular plasmid map rendered from annotated GenBank output	Low		PR Open	It was simple to add with the Rich annotation PR so just went for it.
Import Existing Plasmids	Accept a user-supplied plasmid and modify it per user instructions, detecting existing features first. Use case “take this plasmid and swap out the promoter A for promoter B”	Low			Requires another feature to swap parts

Epic 5: User Experience & Memory

Make the tool feel like a persistent project collaborator, not a one-shot query engine.

Issue	Description	Priority	Assignee	Status	Notes
Project Memory	Persist conversation history across sessions; allow users to return to ongoing projects	Low		Backlog	
Chat memory	The chats are not getting saved.	Low/BUG			
Troubleshooting Mode	Allow users to report experimental failures and receive revised designs with improved reasoning	Low		Backlog	
Custom Lab Rules	Allow labs to define standing preferences (e.g., preferred backbones, promoters, resistance markers) appended to system prompt	Low		Backlog	
Bug Submission Tool	In-app bug reporting with conversation history attached for developer review	Low		Backlog	

Epic 6: Validations

Make the tool feel like a persistent project collaborator, not a one-shot query engine.

Issue	Description	Priority	Assignee	Status	Notes

Validations passing even when construct is wrong	Evaluate the validations to understand how they are occasionally passing when they are wrong	Bug			
Fusion Protein Linker Assessment	Assess whether the linker in a fusion protein is long enough and flexible enough to allow both domains to fold independently. Predict rigid vs. flexible linker behavior.				
Protein Stability Score	Predict the thermodynamic stability (ΔG of folding) of the protein.				
Signal Peptide & Localization Compatibility	Determine whether a signal peptide or targeting sequence will route the protein correctly in a mammalian cell vs. its source organism.				
Post-Translational Modification Sites	Identify glycosylation, phosphorylation, and ubiquitination sites that may not be present or functional if the protein originates from a non-mammalian source.				
Disulfide Bond Compatibility	Flag bacterial proteins that rely on cytoplasmic disulfide bonds, which will fail to form correctly in the reducing environment of the mammalian cytoplasm.				
Proteolytic Cleavage Sites	Identify internal sequences susceptible to cleavage by common mammalian proteases.				
Internal Promoter Conflicts (Lenti)	Flag constructs with two strong promoters in close proximity and the same orientation, CMV usage in silencing-prone contexts, or unintentional promoter elements. Flag any construct with more than two total promoter elements regardless of configuration.				

Cryptic Polyadenylation Signals	Scan the insert for premature polyA signals that would terminate transcription before packaging or expression.				
Cryptic Splice Sites	Flag internal splice donor/acceptor sites that could cause aberrant splicing of the viral RNA during packaging or in the target cell.				
Transgene Biosafety Flags	Warn if the insert encodes a known oncogene, toxin, or otherwise hazardous protein. Note applicable BSL and institutional approval requirements.				
Gain-of-Function Flags	Flag inserts that are constitutively active mutants or engineered to enhance virulence or cell transformation. Require explicit user acknowledgment.				
Total Vector Size	Flag transfer plasmids exceeding ~8–9 kb total. Suggest remediation strategies such as removing WPRE, using a minimal promoter, or splitting the construct.				
Sequence Overlap / Interference Check	Ensure inserts (including promoters, gene of interest, and antibiotic resistance elements) do not conflict with other vectors in the lentiviral series.				
Promoter Compatibility with Target Cell Type	Flag promoters known to be silenced in the specified cell type (e.g., CMV in stem cells or primary cells). Suggest alternatives such as EF1a, PGK, or CAG.				
Pol II vs. Pol III Promoter Selection	Confirm Pol III promoter (U6 or H1) is used for shRNA/sgRNA payloads. Confirm Pol II promoter is used for protein-coding transgenes.				
SIN LTR Confirmation	Confirm the backbone uses a self-inactivating 3' LTR (ΔU3). Flag if absent and require user confirmation that this is intentional.				

WPRE Inclusion Decision	Flag whether WPRE is present. Note that it enhances expression but adds ~600 bp. Prompt user confirmation, especially if the construct is near the size limit.				
Selection Marker Compatibility	Flag selection markers that may have variable or poor efficacy in the specified cell type (e.g., puromycin in certain primary cells).				
Dual Reporter/Selection Cassette Structure	If both a fluorescent reporter and a selection marker are requested, confirm whether the user wants a fusion protein, 2A peptide linkage, IRES, or separate promoters.				
RRE and cPPT Elements Present	Confirm the backbone contains the Rev Response Element (RRE) and central polypurine tract (cPPT). Flag if either is absent.				
Replication-Competent Lentivirus (RCL) Risk	Flag any design choice that reintroduces homology between the transfer and packaging plasmids in a way that increases RCL recombination risk.				
Guide RNA On-Target Efficiency Score	Score guides using established on-target efficiency algorithms (Doench Rule Set 3, DeepCRISPR, or equivalent). Flag guides below acceptable threshold and suggest alternatives. Score at design time and again as part of the overall Design Confidence Score.				
Guide RNA Off-Target Prediction	Run off-target analysis and flag guides with high-similarity matches elsewhere in the genome, particularly those with mismatches only in the PAM-distal region.				
Guide RNA Secondary Structure	Check that the sgRNA spacer sequence does not fold back on itself in a way that would reduce Cas9 loading or activity.				

Guide Targeting Position for Knockouts	For NHEJ-based knockouts, confirm the guide targets an early, constitutively expressed exon and a functionally critical domain. Flag guides hitting late exons or non-essential regions that may produce hypomorphs.				
Multiple Guide Coverage	Recommend designing 3–6 guides per target and validating at least 2–3 experimentally to account for variability in efficiency.				
HDR Donor Template Type	Confirm whether the user is providing an ssODN (suitable for small edits) or a dsDNA template (for larger insertions). Flag size limitations for each.				
Homology Arm Length and GC Content	For ssODN donors, recommend 50–90 bp arms. For dsDNA donors, recommend 400–800 bp arms. Flag arms with very low or very high GC content that may reduce recombination efficiency.				
Distance of Edit from Cut Site	Confirm the desired edit is within 10–20 bp of the cut site. Flag if the edit is far from the cut site and suggest repositioning the guide.				
PAM Disruption in Donor	Confirm the donor template introduces a mutation in the PAM or protospacer to prevent Cas9 from re-cutting after successful HDR. Flag if absent.				
Silent Mutation Strategy	If the donor does not disrupt the PAM, recommend introducing a silent mutation in the protospacer to prevent re-cutting without altering the protein sequence.				
All-in-One Vector Size Check	Flag all-in-one Cas9 + sgRNA constructs that exceed 9–10 kb, as large size reduces transfection efficiency.				
Promoter Selection for Cas9	Confirm the Cas9 promoter is appropriate for the target cell type. Flag CMV usage in stem cells or primary cells where silencing is likely.				

Pol III Promoter for sgRNA	Confirm the sgRNA is driven by a Pol III promoter (U6 or H1). Flag if a Pol II promoter is used unless the user is deliberately using a Pol II-driven guide strategy.				
Antibiotic Resistance Conflicts (CRISPR)	Validate that resistance markers across all plasmids in the system (Cas9 plasmid, donor plasmid, selection cassette) do not conflict with one another.				
CRISPR Selection Strategy	Confirm the user has a plan for enriching edited cells, such as fluorescent sorting, antibiotic selection, or surveyor/sequencing-based screening.				
CRISPR Specificity Score	Report the ratio of on-target to predicted off-target activity. Flag guides with poor specificity ratios.				
Microhomology Prediction	Analyze sequences flanking the cut site for microhomology that would promote MMEJ, producing predictable deletions. Flag as useful or undesirable based on the user's goal.				
HDR vs. NHEJ Bias Prediction	Based on user-provided cell type and cell cycle information, provide a qualitative prediction of whether HDR or NHEJ is more likely to dominate repair outcomes.				
Essential Gene Targeting Flag	Flag if the target gene is essential for cell viability, as efficient cutting may cause cell death rather than a stable knockout line.				
Oncogene / Tumor Suppressor Targeting Flag	Flag edits targeting known oncogenes or tumor suppressors. Note potential effects on cell growth dynamics and interpretation of phenotypic assays.				
Gain-of-Function Edit Flag (CRISPR)	Flag activating mutations being introduced via HDR, particularly if the target gene has known oncogenic potential. Require explicit user				

	acknowledgment.				

Epic 7: Library Generation

Support pooled and arrayed library design for screening applications.

Feature	Description	Priority	Assignee	Status	Notes
Library Spec & Scoping	Define supported library types (sgRNA, ORF, barcode, shRNA) and design requirements	Low		Backlog	
Library Design & Export	Generate library oligo pools compatible with synthesis vendors (Twist, IDT, Agilent)	Low		Backlog	
Parts Library Design	Support construct assembly from modular parts libraries for cell science applications	Low		Backlog	

Epic 8: Security

Empty now, but need to consider how to add layers of security to the platform.

Feature	Description	Priority	Assignee	Status	Notes

Feature Request Notes

Additional Validations Write Up: Link

Requests are not ordered due to prioritization.

Mialy Currently Working On
Good for Anthropic

- Detect MCS within the sequence rather than rely on it being encoded in library metadata (Mialy Currently working on this – Link to WIP PR)
    - This is an issue when a new plasmid backbone is imported and there is no MCS annotation
    - Should be able to rely on the sequence alone to detect a MCS and its orientation.
- Detect the orientation of the plasmid (WIP, will be included in the above PR to detect MCS since this breaks down if the orientation is reversed)
    - Sometimes a promoter is in a reverse (counter clockwise) position on the backbone meaning the gene needs to be reverse transcribed and placed positionally earlier in the sequence
    - The program should be able to detect the MCS and if it's forward or reverse wrt the promoter. It cannot just compare to any promoter since there are usually several promoters within a plasmid.
- Nucleus targeting sequences (retrieve a nuclear targeting sequence)
    - 3xNLS
        - pEGFP-C1 EGFP-3XNLS
        - Prompt: Create a nuclear localized egfp expression plasmid, under control of CMV promoter, for expression in U2OS cells
        - https://www.addgene.org/58468/
        - The program actually does pretty well with this so far. But need to understand why there are difference in designs (Mialy will work on this)
- Improve addgene integration:
    - Get access to single call Addgene API Key. Then allow addgene to be called individually so plasmids can be pulled
    - This feature will require users to get addgene access
    - Doing this instead of a single call once, so that users always have access to the up to date database
- Design Confidence Score: Predict anticipated success of a construct.
    - Link to Document
- Bespoke promoters
    - “Design a p65 promoter reporter construct for stable expression in HEK293 and RAW264.7 cells”
    - It would be ideal if the program understood that this was a custom request, that way it does not get stuck in a loop of trying to look for it (if it does not exist) – in this specific case, there is a p65 promoter reported.
        - Would need to understand if the user was looking for mouse or human p65
    - Before proceeding, could ask the user if they want the program to go into a research mode, if they know the region they want to use as the promoter, or if they just want to take an upstream chunk (1.5-2K bp). Asking first could prevent

a lot of wasted time if the user already knows this has not been done before, or for whatever reason does not want to use a previous construct design.
        - Proceed based on the user input.
- More intelligent fusion protein design (add linkers to disordered loops)
    - Use Case:
        - Fusion protein design is complicated. Sometimes it is simple as adding to the N or C terminal end of the protein, but sometimes adding tags to the ends of proteins will interfere with the target protein function or the fusion protein function. Look through the current protein structure and find the best location to add the linker and fusion protein.
- Smart mutation design
    - Gain of function
    - Loss of function
- Troubleshooting
    - Ideally would retain Project memory
        - Change UI design intention anticipate the user will want to start and returning to “projects”.
        - If a User has started a project and tests the design (in the lab) and it is non functional (say for example a fusion protein design does not work as intended). The user can come back, say what they did and what they observed, and have the program help them troubleshoot.
        - Can provide improved designs and perhaps deeper research.
        - Different codon optimization etc…
- Library Generation
    - Examples of libraries:
        - https://www.addgene.org/pooled-library/
    - Would need to spec this feature out more.
    -
- Improve Insert sequence retrieval.
    - See 10 bug reported above.
    - Incorporate more tooling like Snapgene?
    - Successfully extract genes from plasmid sequences
- Rich plasmid annotation - Right now only annotating the insert. Would be nice to use libraries (can see how Benchling is doing this) or a tool like pLannotate.
- Plasmid visualization

    - Snapshot of the annotated genbank output (First needs rich annotation applied)
    - There must be tools available for this already
- Switch out Ab resistance
    - Use Case:
        - Researchers would usually just move to a different backbone, but since we are just ordering DNA here, we can just switch them out
- Switch out promoters.
    - For this if the user likes all attributes of a backbone but needs a new promoter, would probably be easier to find a new backbone and modify other parts.
    - Do not pick a promoter that is already present within the plasmid.
- Modifying existing plasmids
    - In this case, may not be annotated with all expected attributes.
    - Would need to detect features and modify as the user wants.
    - Might be able to do this already, but need to test.
- Codon optimization
    - Recognize when this is necessary (when switching organisms)
- Output of many sequences
    - Use case prompt:
        - Make me 3 H2B His6 tags with the tags placed in locations that are most likely to not interfere with the folding of H2B or its function. Stable integration in mouse fibroblasts.
- Custom rules consideration.
    - If a lab has a particular way of doing things, allow for them to (via the UI) specify these. This could be appended to the system prompt.
- Take parts of proteins
    - Reference:
        - https://www.molbiolcell.org/doi/10.1091/mbc.e12-09-0685
- Construct plasmids from parts library
    - Feature for Cell Science.
- Add references for decision making (Request from other scientist when presenting some preliminary results)
    - Especially useful once the program starts to do more research
    - Link to PR
- Bug submission tool

    - Retain conversation history and allow user to record the issue that was observed.
    - Will make troubleshooting easier.
    - This project will be in the upcoming Nautalix Hackathon.
    - If we have participants this would be a nice to have to see what users are experiencing
    - I thought I saw logs being saved before but cannot find them now…so many that was for a different project.
- Additional Validations
    - Link to Doc
-